

Contents

Accumulate Project Organization: Final Summary	1
EXECUTIVE SUMMARY	1
DETAILED BRANCH ANALYSIS	1
ISSUES TO BRANCHES MAPPING (101 Active Issues)	4
AIPS: Status and Action Required	4
RECOMMENDED EXECUTION PLAN	5
WORK PRIORITIZATION MATRIX	5
SUCCESS CRITERIA	6
DELIVERABLES CREATED	6
KEY INSIGHTS	7
NEXT IMMEDIATE ACTIONS	7

Accumulate Project Organization: Final Summary

Generated: 2026-04-08

Data Source: Comprehensive branch, issue, and AIP inventory

Analysis: 146 branches, 101+ active issues, complete work mapping

EXECUTIVE SUMMARY

Current State (As of 2026-04-08)

Metric	Value	Assessment
Main Branch	7 commits ahead	Production ready
DAG-BFT Integration	457 commits ahead	Needs stability test
10K TPS Infrastructure	271 commits ahead	Needs validation
Total Branches	146	Needs cleanup
Open Issues	100+	Needs prioritization
AIPs Documented	1 (AIP-006 draft)	Needs definition
Ready-to-Merge	125+ branches	Ready

DETAILED BRANCH ANALYSIS

A. PRODUCTION-READY (Main Branch)

Branch: main

Status: Ready

Details: - 7 commits ahead of origin/main - Includes critical fixes (#3824, #3860, #3866, #3868) - Docker image: accumulated:v1.0.0-critical-fixes (381MB) - Test passed: 882K+ transactions, 100% success rate

Next Steps: - [] Deploy to followers via accman - [] Run 24-hour stability test - [] Validate TPS (7.3K CometBFT baseline)

B. CONSENSUS MODERNIZATION (DAG-BFT)

Branch: dagbft-integration

Status: Feature Complete, Stability Testing Needed

Details: - **457 commits ahead** of main - **Major breaking changes:** Consensus protocol replacement - **Feature Status:** Implementation 80%+ complete - **Performance:** 7.3K TPS measured (same as CometBFT - needs investigation) - **Network Impact:** **Requires complete network reset**

Test Results: - 120-second load test: 7.3K TPS, 100% success - CPU: 42% average, Memory: 23% average - All 12 validators healthy - **Note:** Test may have been DAG-BFT, not CometBFT baseline

Stability Testing TODO: - [] 24-hour stability run (not 120 seconds) - [] Investigate why DAG-BFT TPS not > 7.3K (should be 10K+) - [] Test state consistency across validators - [] Validate snapshot loading - [] Document upgrade procedure - [] Create rollback strategy

Timeline: Ready for testing NOW, deployment in 2-4 weeks

C. PERFORMANCE OPTIMIZATION

Branch: issue-3892-10k-tps-infrastructure

Status: Ready

Details: - 271 commits ahead - 12-node Docker test network - Real-time metrics, monitoring, dashboard - Load testing framework (48 workers) - **Purpose:** Performance baseline and load testing

D. FEATURE BRANCHES WITH REAL WORK (125+ Ready)

TOP PRIORITY - Consensus/Security Fixes By Issue #: | Issue | Branch | Commits | Status | Action | |——-|——-|——-|——-|——-| | #3888 | issue-3888-bpt-parallel-updates | 413 | Ready | Merge to dagbft-integration | | #3869 | Multiple (4 variants) | 12-222 | Ready | Consolidate into single PR | | #3870 | Multiple (4 variants) | 12-225 | Ready | Consolidate into single PR | | #3875 | issue-3875-per-peer-vote-rate-limiting | 231 | Ready | Review & merge | | #3873 | issue-3873-optimize-lru-eviction-locking | 224 | Ready | Review & merge | | #3826 | issue/dagbft-3826 | 10 | Ready | Merge to dagbft-integration | | #3825 | issue/dagbft-3825 | 12 | Ready | Merge to both branches |

HIGH PRIORITY - DAG-BFT Service Architecture By Issue # (3800-3830): | Issue | Branch | Commits | Status | |——-|——-|——-|——-| | #3823 | issue/dagbft-3823 | 25 | Ready | | #3817 | issue/dagbft-3817 | 12 | Ready | | #3815 | issue/dagbft-3815 | 12 | Ready | | #3802 | issue/dagbft-3802 | 12 | Ready | | ... | issue/dagbft-3816 through 3830 | 10-25 | Ready |

INFRASTRUCTURE/TESTING

Issue	Branch	Commits	Status
#3854	feature/issue-3854	10	Ready
#3842-3848	issue-3842+	172-177	Ready
#3839-3841	issue-3839+	172-174	Ready

FEATURE WORK (Other Real Implementations)

Issue	Branch	Commits	Last Activity	Status
#3862	feature/issue-3862	4454	2026-03-27	Needs rebase
#3863	feature/issue-3863	215	2026-03-23	Ready
#3857-3858	feature/issue-38XX	1-6	2026-03-23	Ready
#3824, #3843-3847	feature/issue-38XX	1-2	2026-03-22	Ready
#3713, #3705	3713-add-version, 3705-whitelist	6-9	2026-03-26	Ready

E. STALE / DEAD BRANCHES (Cleanup Required)

Dead Branches (Archive Immediately)

Branch	Commits	Status	Action
feature/issue-3862	4454	Dead	Delete - needs rebase to current base
issue-3877-replace-math-rand	4453	Dead	Delete
issue-3876-race-conditions	4452	Dead	Delete
issue-3801-dagbft-validator	4451	Dead	Delete

Reason: Heavily diverged (4000+ commits ahead), rebasing not feasible

Stale/Dormant Branches (Reassess) Mining-related (AIP-53): - fix/dennis-mining-correct-crypto (286 commits, 2026-03-26) - 3640-mining-support (5+ months old) - 3665/3680 lxr-mining branches

Status: Blocked pending AIP-53 decision

ISSUES TO BRANCHES MAPPING (101 Active Issues)

CRITICAL ISSUES

Issue	Title	Branch(es)	Status	Action
#3888	BPT Parallel Updates	issue-3888-bpt-parallel-updates (413c)	Ready	Merge
#3869	Vote Spam Fix	4 branches (222c)	Ready	Consolidate
#3870	Vote CPU Fix	4 branches (221c)	Ready	Consolidate
#3892	10K TPS Infrastructure	issue-3892-10k-tps (271c)	Ready	Validate
#3826	HTTP 429 Backpressure	issue/dagbft-3826 (10c)	Ready	Merge
#3825	Prometheus Metrics	issue/dagbft-3825 (12c)	Ready	Merge

BLOCKING DEPENDENCIES

1. **#3888** → **dagbft-integration** (BPT sharding)
2. **#3869/#3870** → **dagbft-integration** (Consensus security)
3. **#3873/#3875** → **dagbft-integration** (Rate limiting/LRU)
4. **#3892** → **Testing** (10K TPS validation)

AIPS: Status and Action Required

AIP-006

- **Status:** Draft (found in test code)
- **Location:** `/test/e2e/sig_general_test.go`
- **Action:** Define scope and requirements

AIP-53 (Mining)

- **Status:** BLOCKED (5+ months dormant)
- **Related Issues:** #3885 (URGENT flag)
- **Related Branches:** 7 dormant branches
- **Action:** **URGENT** - Schedule decision (continue/pause/cancel)

AIP-54

- **Status:** Referenced in #3886, no definition
- **Action:** Define requirements

Other AIPs

- **Status:** Not documented in code
- **Likely Location:** External (GitLab wiki, documentation, design docs)
- **Action:** Centralize AIP documentation

RECOMMENDED EXECUTION PLAN

PHASE 1: IMMEDIATE (This Week)

Priority 1a: Critical Decisions - [] Schedule AIP-53 decision (1-2 hours) - Continue: Consolidate 7 branches - Pause: Archive branches - Cancel: Close related issues - [] ARM64 crypto assessment (#3884) (4-8 hours)

Priority 1b: Branch Consolidation - [] Consolidate 4-branch duplicates (#3869/#3870) into single PR - [] Clean up dead branches (4000+ commits) - [] Rebase stale feature branches

PHASE 2: SHORT TERM (2-4 Weeks)

Priority 2a: DAG-BFT Validation - [] Run 24-hour stability test on dagbft-integration - [] Investigate TPS plateau at 7.3K (should be 10K+) - [] Test state consistency - [] Document upgrade procedure

Priority 2b: Merge Ready Work - [] Merge #3869/#3870 consolidated PR - [] Merge #3875 vote rate limiting - [] Merge #3873 LRU optimization - [] Merge #3826/#3825 service APIs

Priority 2c: Issue Prioritization - [] Triage all 100+ issues - [] Assign priority (critical/high/medium/low) - [] Create roadmap

PHASE 3: MEDIUM TERM (1-3 Months)

Priority 3a: Process Implementation - [] Create branch naming policy - [] Implement 1-branch-per-issue workflow - [] CI/CD automation for branch cleanup

Priority 3b: Production Deployment - [] Deploy v1.0.0-critical-fixes to followers - [] Validate 24-hour stability on production - [] Plan DAG-BFT network upgrade

Priority 3c: Feature Integrations - [] Merge other feature branches as ready - [] Integration testing across features - [] Release planning

WORK PRIORITIZATION MATRIX

Must Do (Next 2 Weeks)

1. AIP-53 decision - **impacts 7 branches**
2. DAG-BFT stability test - **production-critical**
3. Consolidate duplicate PRs - **10+ duplicate commits**
4. Deploy v1.0.0 to followers - **production readiness**

Should Do (Next 4 Weeks)

1. Merge consensus/security fixes
2. Triage and plan 100+ issues
3. Implement branch policy
4. Integration test feature branches

Nice to Have (Next 3 Months)

1. Consolidate other feature work
 2. Complete remaining optimizations
 3. Full AIP documentation
 4. Architecture documentation
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SUCCESS CRITERIA

By End of Week

- ☐ AIP-53 decision made
- ☐ Dead branches deleted
- ☐ Duplicate PRs consolidated
- ☐ v1.0.0 deployed to 1+ follower

By End of Month

- ☐ DAG-BFT stability validated (24 hours)
- ☐ 5+ major PRs merged
- ☐ All 100+ issues triaged
- ☐ Branch policy implemented

By End of Quarter

- ☐ DAG-BFT upgrade date scheduled
 - ☐ All feature branches merged
 - ☐ Roadmap execution started
 - ☐ <30 active branches (cleanup complete)
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DELIVERABLES CREATED

Analysis Documents

1. **ACCUMULATE_ORGANIZATION.md** - Full inventory with 6 sections
2. **IMMEDIATE_ACTION_ITEMS.md** - Prioritized work items
3. **AIP_ISSUE_BRANCH_MAPPING.md** - Traceability matrix
4. **FINAL_ORGANIZATION_SUMMARY.md** - This document

Data Files (CSV)

1. **INVENTORY_01_BRANCHES.csv** - 146 branches with metadata
2. **INVENTORY_02_ACTIVE_ISSUES.csv** - 101 active issues
3. **INVENTORY_03_AIPS.csv** - AIP documentation

Ready to Use

- Import CSVs into Excel/Sheets for filtering

- Use markdown files for strategy discussions
 - Reference action items for sprint planning
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KEY INSIGHTS

What's Working

Main branch clean and production-ready
125+ branches ready to merge
Strong test infrastructure (12-node network)
Clear issue tracking (100+ documented)

What Needs Attention

Branch sprawl (146 total, many duplicates)
Stale work (4000+ commit branches, 5-month dormancy)
AIP documentation missing (mostly external)
No clear prioritization (100+ issues, no roadmap)
Duplicate effort (4-branch variants for same issue)

What's Blocked

AIP-53 Mining (decision needed)
DAG-BFT upgrade (stability testing)
Feature consolidation (process clarity)

NEXT IMMEDIATE ACTIONS

Do Today: 1. Schedule AIP-53 decision meeting 2. Triage #3884 ARM64 crypto issue 3. Assign owners to critical items

Do This Week: 1. Delete 4 dead branches 2. Consolidate #3869/#3870 PRs 3. Start DAG-BFT 24-hour test

Do Next 2 Weeks: 1. Deploy v1.0.0 to followers 2. Merge consolidated security PRs 3. Complete issue prioritization